

Reading Line Classification Using Eye-Trackers

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Abstract—Eye-tracking while reading is an emerging application where the goal is to track the progression of reading. The challenges for accurate tracking of the reading progression are due to the measurement noise of the eye-tracker and the rapid and uncertain movement of the eye gaze. Solutions to this problem developed in the recent past suffer from many limitations, such as the need to know the text context and the need to have a batch of one page of data for classification. In this article, we relax these assumptions and develop a novel, real-time line classification approach. The proposed solution consists of an improved slip-Kalman smoother (slip-KS) that is designed to detect new line returns and to reduce the variance in the eye-gaze measurements. After preprocessing of the data by the slip-KS, a classification approach is employed to track the lines being read in real-time. Two such classifiers are demonstrated in this article; one is based on Gaussian discriminants, and the other is based on support vector machines. The proposed approaches were tested using realistic eye-gaze data from seven participants. Analysis based on the collected data using the proposed algorithms shows significantly improved performance over existing methods.

Index Terms—Classification, eye-gaze fixations, eye-tracking, Gaussian discriminants, human-computer interaction, Kalman filter, Kalman smoother (KS).

I. INTRODUCTION

EYE-TRACKERS are emerging as a new form of measuring systems that estimate several physiological features of a person, such as eye-gaze position and pupil diameter [1]. Eye-trackers used to be costly and bulky with limited applications in laboratories. With recent advances in video sensors and embedded computing systems, low-cost, desktop-mounted eye-trackers are being experimented for a wide variety of applications. However, improving the accuracy and performance of desktop-mounted eye-trackers is an active research domain. In [2], a video-based eye gaze tracking method was proposed to precisely detect the corneal reflection of light and pupil. In order to get more reliable measurements, the authors [3]–[5] provide several solutions under the scope of calibration. In addition, head movement detection system is proposed in [6] in order to eliminate the effects of head movement and improve eye-tracking accuracy.

With the development of low cost eye-tracking technology [7], research into eye-tracking has become commonplace

in wide range of applications, such as advanced driver assistance systems [8], human computer automation [9], robotics [10], market research [11], eye gestures detection [12], and psychology research [13]. With the advances in sensing technology, computing, connectivity, and storage, it is now possible to collect eye-tracking data at negligible cost in a wide variety of situations. Research into eye-tracking has seen exponential growth in the past two decades. In the same period, the accuracy of low-cost eye-trackers has also seen significant improvement [7]. Reading line classification is one of the futuristic research areas of eye-tracking that has applications in such fields as reading comprehension analysis [14], reading interest analysis [15], and online education [16].

Typical eye-tracker measures several parameters of the eye and its movement, such as gaze locations, gaze fixations, pupil parameters, and eye-blink information. Particularly, eye-gaze measurements [17] are of interest in this article. Eye-gaze points have information about the mental state of the subject [18] as well as it contains information about the visual interaction of the subject with their surrounding environment [19]. Using the eye-tracker data, it is possible to know more about the interests and experience of the subject, e.g., interests while reading a book. However, there are significant challenges faced by today's eye-trackers—particularly when it comes to tracking the progression of reading—from the fact that the low-cost eye-trackers have meager resolution compared to what is required. Consequently, there is significant research interest related to analyzing eye-tracking while reading. In Section II, we will briefly go through these recent studies.

The purpose of this article is to track the progression of reading, especially the line being read, utilizing the data collected by the Gazeport GP3 desktop mounted eye-tracker [20] when participants read text on a computer screen without any prior knowledge about the content. The eventual goal of developing this technology is to help understand the level of interaction a person had with a text passage. Such technology will be useful to classify the reading pattern into known categories [21], e.g., reading, skimming, and scanning. Furthermore, an accurate reading line classification algorithm can have more potential applications, such as online training and education, retail research, marketing, psychological studies, cognitive research, and testing of reading comprehension [22]. It is also important to note that some languages are read from left to right while others are read from right to left. The line classification algorithm presented in this article is applicable to work under both scenarios.

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The contributions of this article are as follows.

- 1) *Slip-Kalman Smoother (slip-KS) Approach* to preprocess the eye-gaze data so as to reduce the measurement noise and to decouple the gaze data corresponding to different lines. The proposed approach is found to improve the line detection performance compared to a recent prior work reported in [23] and [24].
- 2) *Application of Gaussian Discriminant Classifier for Line Detection While Reading*: This article introduces the Gaussian discriminant approach, for the first time, to line detection problem. A gaussian discriminant is a robust approach to classification. However, eye-gaze data, while reading, are scattered across the page, partly due to the nature of the eye-gaze and partly due to the measurement noise of the eye-tracker. The preprocessing approach in this article, using the KS, enabled effective application of the Gaussian discriminant classifier and resulted in a robust and computationally efficient approach to the line detection system.
- 3) *Novel Feature Selection Strategy for line Detection Algorithms*: The new feature selection strategy was tested with the support vector machine (SVM) as the classifier. The new strategy used information from both x - and y -coordinates, and selected feature evenly from the entire line. The resulting performance is found to be superior to existing approaches.
- 4) *Validation Using Realistic Data*: The performance of the proposed approaches was evaluated using data collected from seven participants.

The proposed approaches in this article are significant for the following reasons.

- 1) The commercial (particularly low-cost) eye-trackers come with significant measurement noise [25] both due to limitations of the sensors as well as the limitations of the state-of-the-art eye-gaze processing algorithms. The proposed approaches in this article could be used by future eye-tracker developers to improve their outputs. Also, the proposed approaches can be useful in webcam-based eye-gaze tracking applications [26] where the measurement noise is significant.
- 2) When a person reads, the eye-gaze might wander into the previous or subsequent lines—this does not mean the person switched lines; rather, it is a feature of the human gaze; this feature needs to be filtered out to get the correct picture of the reading pattern. The proposed approaches in this article are designed to address this.
- 3) Previous research [23] on the same problem assumed that the number of lines on a page and the coordinate of each line are known as prior knowledge. These two assumptions are relaxed in this article, i.e., the numbers of lines and their locations are learned during the training phase, which means that the lines of text do not need to be equally spaced.

The remainder of this article is organized as follows. Section II briefly introduces recent studies about eye-gaze tracking analysis while reading. Section III describes the proposed approach in this article. Section IV presents the

results of the proposed approach using real-world data. Finally, this article is concluded in Section V.

II. RELATED WORK

In this section, a review of recent studies on reading procedure analysis via eye-tracking technology is presented.

A. Applications of Eye-Tracking While Reading

The main research area about eye-tracking analysis while reading is reading comprehension. Copeland *et al.* [27] proposed a new performance function for training feed-forward neural networks to predict reading comprehension based on eye gaze data. Also, Sanches *et al.* [28] extracted some specific eye-gaze features and used a support vector regression model to predict participants' percentage of correct answers about the contents in the text. Similarly, another approach measures participants' comprehension ability by calculating reader ranking based on correct answer lines recorded by the eye gaze tracker and the number of correct answers given by participants [29]. In addition to reading comprehension, some other research domains also received significant attention to eye-tracking while reading. In [30], supervised machine learning techniques were used to detect mind wandering probes from eye gaze and context features to analyze which features had the most significant impact. Moreover, Lara-Alvarez *et al.* [31] introduced an approach called *counting words from lines (COWL)*, which deals with the imprecision problem by associating the eye-tracking data with points obtained from character recognition.

Various research applications of eye-tracking while reading are still at the early stage due to the fact that accurate enough eye-gaze tracking—that is sensitive enough to detect the line being read—still requires separate hardware. This might rapidly change when the eye-gaze tracking technology based on built-in webcams matures [32]. Most present studies directly use the raw eye-gaze data from the eye-tracker, which has noise and disturbance caused by the device and participants [33]. As opposed to many existing approaches that use raw eye-gaze data to extract statistical information of the reading behavior, this article focuses on improving the eye-tracking technology itself while reading.

B. Tracking Accuracy Improvement

Only a few prior works proposed preprocessing techniques to eye-gaze data. In [34], an approach was proposed to correct the vertical error of the gaze fixations and alignment of estimated eye-gazes with text resulting in improved accuracy of line classification. However, this approach only considered fixations and vertical error—in the eye gaze context, the gaze position's distribution has more information. In a recent work, Bottos *et al.* proposed a hidden Markov model (HMM) based line detection system (LDS) [23]. The proposed LDS was demonstrated using data collected from one participant where the lines of text spanned the entire page (page width), and the reading always occurred line-by-line without any repetitive reading of lines. The proposed HMM-based LDS also assumed

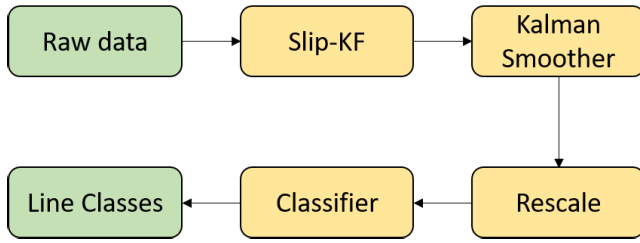


Fig. 1. Procedure of whole experiment. This is the diagram of the whole classification approach.

the number of lines as well as their locations on a page. In order to be widely applicable, an LDS needs to be able to track without requiring to know the number of lines on the screen and their exact locations, because a user can easily move the location of a text (e.g., an e-book) on a computer screen.

III. PROPOSED APPROACH

In this section, a new approach for real-time line classification is presented. Fig. 1 shows the workflow diagram of the proposed approach. First, the raw eye-gaze data is sent through the slip-Kalman filter (slip-KF) [23] which separates data corresponding to different lines. The decouple data, i.e., the data belonging to each line is then sent through a KS, which is employed to reduce the variance in the filtered data. Hereafter, the above two steps, the slip-KF followed by the KS, is referred to as the slip-KS. The slip-KS output is then rescaled to span the entire text region. After the above preprocessing, data from few pages are used to train the classifiers. The parameters from the classifier output are used for line classification.

The remainder of this section will detail each block in Fig. 1.

A. Regular Kalman Filter

Let us consider the following 4×1 state vector:

$$\mathbf{x}(k) = [x(k) \quad \dot{x}(k) \quad y(k) \quad \dot{y}(k)]^T \quad (1)$$

where $x(k)$ is the x -coordinate of the eye-gaze at time step k , $y(k)$ is the y -coordinate of the eye-gaze at time step k , $\dot{x}(k)$ is the velocity of the eye-gaze movement in the x -direction at time step k , and $\dot{y}(k)$ is the velocity of the eye-gaze movement in the y -direction at time step k . The *process model* [35], based on the state vector introduced in (1) is written as

$$\mathbf{x}(k+1) = \mathbf{F}\mathbf{x}(k) + \mathbf{\Gamma}\mathbf{v}(k) \quad (2)$$

where vector $\mathbf{v}(k)$ is a 4×1 zero-mean Gaussian noise with unity standard deviation,

$$\mathbf{F} = \begin{bmatrix} 1 & T & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \Delta T \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \mathbf{\Gamma} = \begin{bmatrix} \frac{1}{2}\Delta T^2 & 0 \\ \Delta T & 0 \\ 0 & \frac{1}{2}\Delta T^2 \\ 0 & \Delta T \end{bmatrix} \begin{bmatrix} q_x & 0 \\ 0 & q_y \end{bmatrix} \quad (3)$$

q_x and q_y are the parameters of the process noise covariance $\mathbf{Q}$, which can be shown as [35]

$$\mathbf{Q} = E\{\mathbf{\Gamma}\mathbf{v}(k)\mathbf{v}(k)^T\mathbf{\Gamma}^T\} \quad (4)$$

$$= \begin{bmatrix} \frac{1}{4}\Delta T^4 q_x & \frac{1}{2}\Delta T^3 q_x & 0 & 0 \\ \frac{1}{2}\Delta T^3 q_x & \Delta T^2 q_x & 0 & 0 \\ 0 & 0 & \frac{1}{4}\Delta T^4 q_y & \frac{1}{2}\Delta T^3 q_y \\ 0 & 0 & \frac{1}{2}\Delta T^3 q_y & \Delta T^2 q_y \end{bmatrix}. \quad (5)$$

After the *process model*, the *measurement model* of reading can be built as follows:

$$\mathbf{z}(k) = \mathbf{H}\mathbf{x}(k) + \mathbf{w}(k) \quad (6)$$

where

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad (7)$$

and

$$\mathbf{z}(k) = [z_x(k) \quad z_y(k)]^T \quad (8)$$

is the eye-gaze measurements, which are the noisy observations of $x(k)$ and $y(k)$. The 2×1 vector $\mathbf{w}(k)$ indicates measurement noise.

After the definition of the state-space model comprising of (2) and (6), the KF can be summarized through the diagram shown in Fig. 2, where $\hat{\mathbf{x}}(k|k)$ is the estimated state vector at time k and $\mathbf{P}(k|k)$ is the state covariance matrix ($k = 1, \dots, N$, where N is the total number of gaze points). For more details of the parameters and initializations refer to [24].

Fig. 3 shows the performance of the regular KF, where the red dots indicates the measurements $\mathbf{z}(k)$ and the blue lines are the estimate from KF. From Fig. 3, it can be seen that the KF was able to follow the reading progress. However, the KF estimates overshoot when the reading progress returns to a new line—this can be used to detect line returns [24].

B. Slip Kalman Smoother

It was demonstrated in [24] that the slip-KF can be used to decouple the data belonging to each line and remove the overshoot. The slip-KF is inspired by slip gear [36], detecting the reading line return when a specific threshold is exceeded. So, the problem transfers into finding a proper indicator to detect the line return while reading.

From Fig. 4, the dramatic change in velocity of the x -direction $\dot{x}(k)$ can be clearly seen. As a consequence, the slip-KF chooses $\dot{x}(k)$ to be the indicator of line return. The final slip-KF can be summarized as follows: When running the regular KF, detect the unusual changes in estimated velocity $\hat{\dot{x}}(k)$ in time step k . If $\hat{\dot{x}}(k) < v_{th}$, reinitialize the state vector $\hat{\mathbf{x}}(k+1|k+1)$ of next time step $k+1$

$$\hat{\mathbf{x}}(k+1|k+1) = [z_x(k+1), v_0, z_y(k+1), \hat{\dot{y}}(k|k)]^T$$

where v_{th} is a threshold on the x -velocity, and v_0 is the initial x -velocity that usually is a predetermined value for different

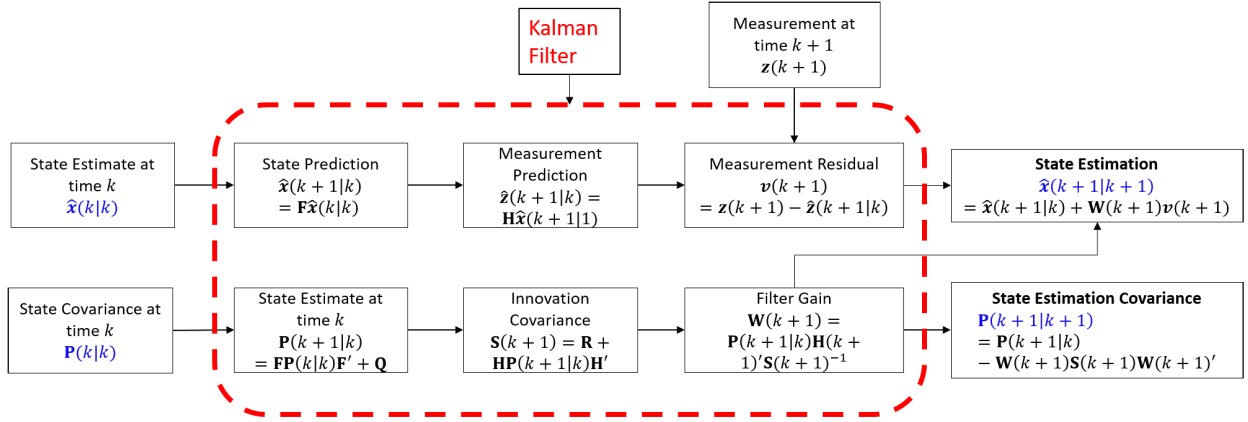


Fig. 2. Regular Kalman filter. Regular KF can be used to filter eye-gaze data by selecting appropriate values for the process and measurement noise as it is described in Section III-A. However, provided that the eye-gaze is engaged in reading, a slip-KF (see Fig. 5) will improve the eye-gaze tracking performance while reading.

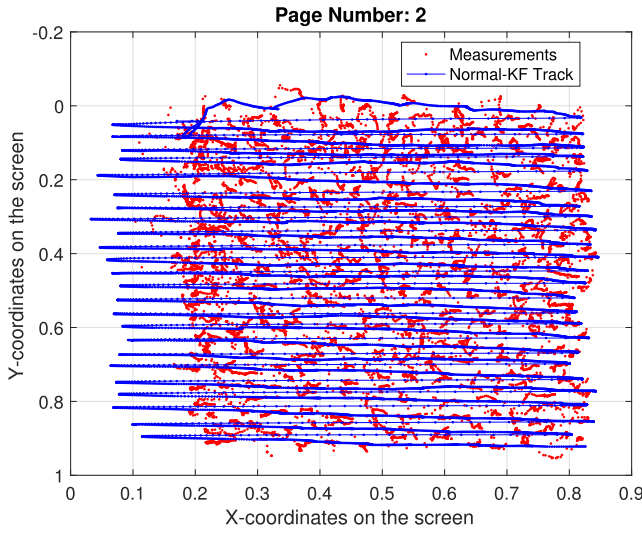


Fig. 3. Tracking eye-gaze while reading using regular Kalman filter. The problem with the regular KF is illustrated as the backtrack over-estimation on the left.

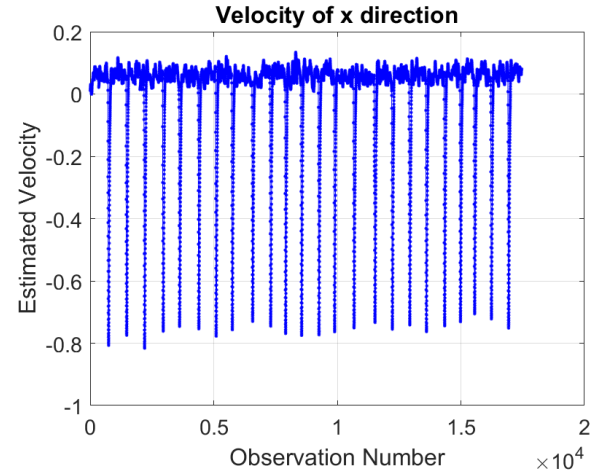


Fig. 4. Velocity $\dot{x}(k)$ (text-width)/sec of regular KF. Velocity of the x-direction is a good indicator since the line return can be distinguished very clearly when the velocity goes to a relatively large negative value.

types of reading. The state covariance matrix $\mathbf{P}(k+1|k+1)$ re-initializes to the initial state $\mathbf{P}(0|0)$. The remaining operations of the slip-KF are the same as the regular KF. Fig. 5 shows a block diagram of one recursive step of the slip-KF.

The slip-KF approach described above is re-initialized at the start of every new line to preset values. This would eliminate the over-shot estimates shown in Fig. 3. However, the output of the slip-KF is still observed to contain high variance. Thus, the slip-KS is introduced to further reduce the variance in the estimates. After slip Kalman filtering, the entire eye-gaze data is decoupled into different lines. The slip-KF estimates (i.e., $\hat{x}(k|k)$ and the covariance matrix $\mathbf{P}(k|k)$) corresponding to one line are then sent through a KS. This process comprises the slip-KS step. For more details about KS, the reader is referred to [35].

Fig. 6 shows the output of KS compared to that of the KF for few lines; the smoothing results of the KS can be clearly observed in this figure. Fig. 7 shows the performance comparison between the two filters more clearly; the KS provides smoother gaze position estimates compared to the

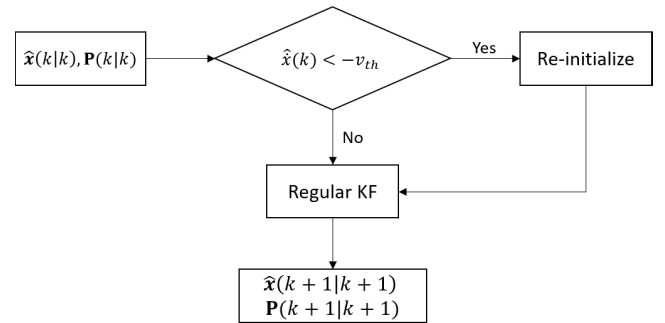


Fig. 5. Slip Kalman filter. The slip-KF is suitable for eye-gaze tracking while reading. v_{th} is the general expression of the velocity threshold.

KF. Also, as shown in Fig. 8, the smoothed data from KS distinguishes the lines much more clearly.

C. Classification Approach

In this subsection, we present the line classification algorithm. We tested two different classifiers: one is the

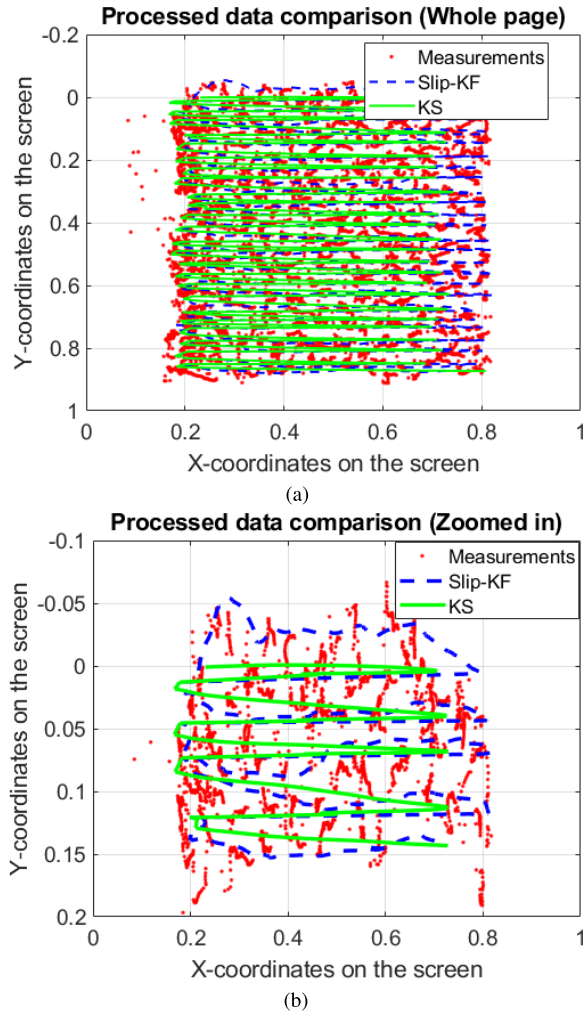


Fig. 6. Eye-gaze tracking while reading smoothed by KS. The previous slip-KF is able to follow the eye-gaze measurements along the line. However, the y-coordinate still has some disturbance which will have influence on the proposed classification algorithm. In this figure, the KS can eliminate more noise on y-coordinate than the previous slip-KF. (a) Entire page. (b) Zoomed-in version.

proposed Gaussian discriminant classifier, and the other is an SVM-based classifier. We will discuss each of these approaches in the remaining part of this subsection.

1) *Gaussian Discriminant Classification*: Fig. 8(a) shows the y-coordinates of the gaze measurements as well as the estimated y-coordinates using the slip-KS in the form of a histogram. The data corresponding to each line is shown in different colors. The leftmost histogram corresponds to the top line (line-1) and the rightmost histogram corresponds to the line at the bottom (line-25). It can be observed that the slip-KS outputs are much more concentrated in each line than the raw observations. In other words, the slip-KS outputs are separated well according to each line.

Another observation from Fig. 8(a) is that the y-coordinates are distributed according to a Gaussian-like distribution. This observation is then used to approximate data corresponding to each line as a Gaussian distribution. Fig. 8(b) shows the Gaussians computed for each line from page 23 of the

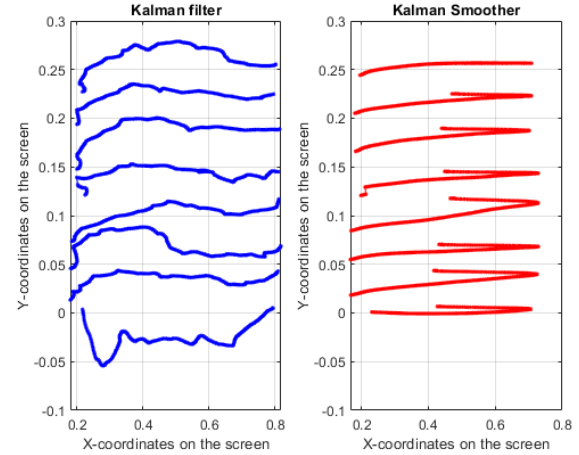


Fig. 7. Comparison between KF and KS results. In contrast, KS can more effectively reduce the noise than KF. After smoothing, the process of reading becomes clearer.

collected data—the top-plot shows the Gaussian distribution with parameters computed from the raw data, and the second plot shows the same with parameters computed from the Kalman smoothed data. Similar to Fig. 8(a), the leftmost PDF corresponds to the top line (line-1) and the right most PDF corresponds to the line at the bottom (line-25) in the Fig. 8(b). The parameters of this Gaussian distribution will be used in the classification algorithm.

In order to evaluate our proposed approach, without loss of generality, the leave-one-out cross-validation (LOOCV) [37] approach was used. Assume we had N pages of eye-gaze data while reading, then extract page j where $j \in \{1, 2, \dots, N\}$ to be the test set and pages $1, 2, \dots, j-1, j+1, \dots, N$ were used as the training set. With this assumption, the parameter of line i can be written as (μ_i, σ_i) where μ_i is the mean of the line i y-coordinates from pages $1, 2, \dots, j-1, j+1, \dots, N$ and σ_i is the standard deviation of line i 's y-coordinates from pages $1, 2, \dots, j-1, j+1, \dots, N$.

Now, once all N sets of parameters $(\mu_i, \sigma_i, i = 1, 2, \dots, 25)$ are computed it is ready to classify the lines in page j . The objective of the classification is as follows: given the y-coordinate of a gaze point $\hat{y}(k|n)$ the log-likelihood that the observation might have been generated from line i is computed as follows:

$$\begin{aligned} d_i &= \log \left(\frac{1}{\sqrt{2\pi}\sigma_i} e^{-\frac{(\hat{y}(k|n) - \mu_i)^2}{2\sigma_i^2}} \right) \\ &= \log \left(\frac{1}{\sqrt{2\pi}\sigma_i} \right) - \frac{(\hat{y}(k|n) - \mu_i)^2}{2\sigma_i^2} \end{aligned} \quad (9)$$

where d_i denotes the likelihood that the gaze data $\hat{y}(k|n)$ comes from line i . The estimated line corresponding to the k th observation is then given as

$$\text{class}(k) = \arg \max_i d_i \quad i = 1, 2, \dots, 25. \quad (10)$$

In other words, this article transfers this problem into a multiclass Gaussian discriminant analysis [38]. Algorithm 1 is how the classification approach, log-likelihood (LLH) classifier, works. However, due to the differences in the head position of

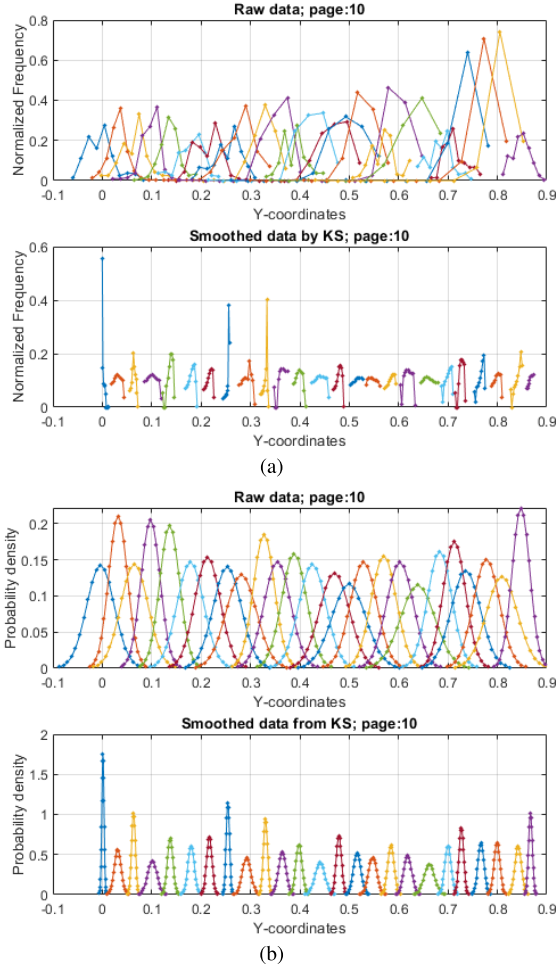


Fig. 8. Distribution of y-coordinate. (a) and (b) Distribution of the eye-gaze measurements (y-coordinates) while reading one page containing 25 lines. (a) Histogram of one page. (b) Gaussian distribution of one page.

participants and other external factors of the eye tracker, each page's distribution is not the same. For instance, the mean of different pages' first line is different, and the ranges of different pages are also different. In order to overcome this distribution, the rescale approach plays a vital role. Before the classification, the y-coordinate of every page has to be rescaled into the same y-coordinate region, which is $[0, 1]$, for data preprocessing.

Algorithm 1 (Gaussian-Based Classifier)

Input: N pages training data

Output: estimated line **class** of test data

- 1: Select page/participant j to be the test set and others be the training set. ($j = 1, \dots, N$)
 - 2: Use the training data to calculate the mean y-coordiante $\mu(i)$ and standard deviation $\sigma(i)$ of every line
 - 3: **for** $k = 1:\text{length}(\text{test data})$ **do**
 $\text{LLH}(k) = -((\hat{y}(k) - \mu)^2 / 2\sigma^2)$
 $[\sim, \text{class}(k)] = \max(\text{LLH}(k))$
 - 4: **end for**
-

The classification approach, explained so far for each point, can be simply extended to a *line-by-line classification* approach as follows. Since the slip-KF records the line return while reading, it is easy to split the measurements into different lines. After splitting the measurements, all the measurements are labeled with a pre-estimated line number. Then the Gaussian-based classifier is used to predict the line classes for every measurement and a majority voting is applied to assign a line number for each separated line.

2) *SVM Classification With Novel Feature Selection*: The second type of classifier is implemented using SVM. A built-in multiclass-SVM classifier available in MATLAB is used for this. The performance of a classifier depends on the selection of features. In this subsection, we describe a feature selection strategy for improved classification performance.

The SVM classifier requires an equal number of data points (features) from each line, here, the line number serves as the class label, i.e., when there are 25 lines on each page, there are 25 possible classes. The features could be the y-coordinates from the slip-KS output, i.e., $\hat{y}(k|n)$. Using $\hat{y}(k|n)$ as a feature is not feasible due to the fact that n varies for each line. In order to obtain an equal number of features that are most representative of each line, the slip-KS estimates of the x-coordinates are used as follows. $\hat{x}(k|n)$ is divided into 40 bins of equal width between $\hat{x}(1|n)$ and $\hat{x}(n|n)$; then, ten $\hat{y}(k|n)$ samples are randomly selected from each bin; this resulted in 400 features for each line. If a certain bin had less than ten samples, then the required number of samples for that bin was taken from adjacent bins by making sure that samples were not duplicated. One clear advantage of the proposed feature selection strategy is that the information from the x-coordinate is explicitly used to select features. The SVM-based approach also employs LOOCV to guarantee generality.

IV. EVALUATION AND RESULTS

A. Data Collection Procedures

In this section, two datasets are used to evaluate our proposed algorithm. First, we use the data from [23] to demonstrate the proposed method, which has 25 pages of data from one participant (a male in his twenties). Eye gaze data were collected utilizing Gazepoint GP3 [20] a desktop mounted eye-tracker at a sampling rate of 60 Hz. As shown in Fig. 9 (left), the eye-tracker is placed at the bottom of the computer screen (a 1920×1080 PC screen), and the recorded position of eye-gaze data while a person reads one page of text consisting of 25 lines is shown on the right. After the eye-tracker was set up, the participant sat in front of the screen and went through a calibration process for the eye-tracker. The eye-tracker software offers a 9-point calibration and 5-point one; the participant was asked to select the former. During the 9-point calibration, the eye-tracker interface displays bright circle markers at nine random positions on the screen and the participant was asked to look at them. If the calibration fails (this happens when the participant does not track the displayed markers), the participant will be prompted to redo the calibration step. Once the calibration is done, the participant remained seated at the same position and

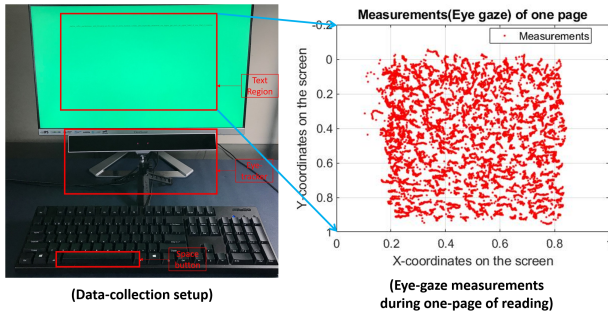


Fig. 9. Experiment Setup. Participants will sit in front of the monitor where the text for reading will be displayed in full-screen mode (left). This geometry causes a vertical measurement bias that is discussed in Section IV. The distribution of one page collected eye-gaze measurements (right).

proceed to the reading step. During the reading, the participant was required to press the space key to reveal a single text line. Each keystroke will only show one line, starting from line 1 at the top of the screen and ending at line 25, and will hide the previous line. In this way, every time the participant finished reading one line, he pressed the space key to display the next line—this allowed to record the ground truth. The x - and y -coordinate of the eye-gaze measurements will be used as the input raw data to do the analysis.

In order to further validate the algorithms, we collected more data from six additional participants in the same experimental setup as above. This time, the participants read only ten pages. Then, we combined these six data with the first ten pages of the prior data. Eventually, we had two datasets, the first dataset had one participant reading 25 pages and the second one had seven participants reading ten pages each, respectively.

B. Evaluation Metrics

The classification error for page p is defined as

$$e_p = \left[\frac{\text{count}_{t=1}^{t=N_p} (\hat{S}_p(t) \neq S_p(t))}{N_p} \right] \times 100 \quad (11)$$

where $\hat{S}_p(t)$ denotes the estimated line number, $S_p(t)$ denotes the true line number (ground truth), and N_p denotes the number of data points from page p . The average error across all pages, e_{avg} was then computed as

$$e_{\text{avg}} = \frac{\sum_{p=1}^N e_p}{N} \quad (12)$$

where N denotes the number of pages. The classification accuracy is then defined as

$$\text{Line Detection Accuracy (in \%)} = 100 - e_p. \quad (13)$$

In the remainder of this section, the classification accuracy of various approaches is presented.

C. Classification Results

Fig. 10(a) shows the prediction accuracy of the proposed classification approach. The line detection accuracy is computed by averaging the percentage of correct classification

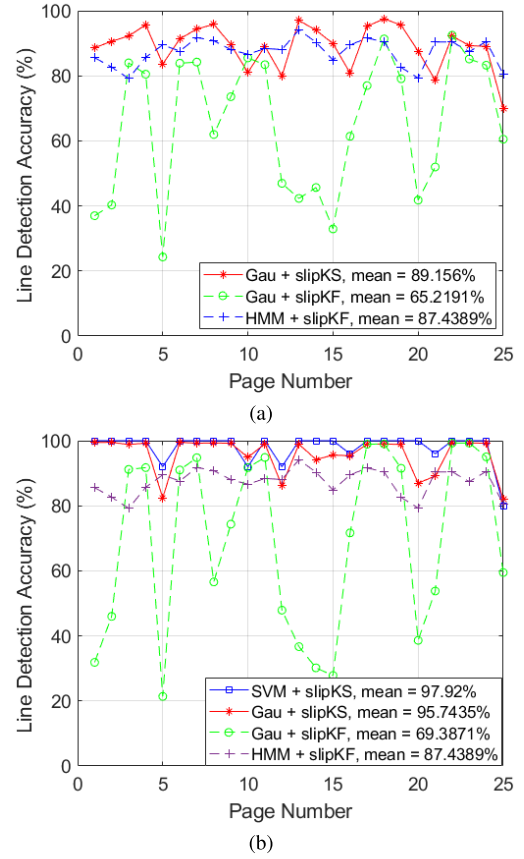


Fig. 10. Classification Accuracy. In (a), the proposed point-by-point (or measurement-by-measurement) line classification is compared with that of the HMM approach in [23]. In (b), different line-by-line classification accuracies are compared. (a) Point-by-point accuracy. (b) Line-by-line accuracy.

from all 25 pages. For comparison, the line detection from slip-KF based estimates and slip-KS based estimates are shown in the same plot. Furthermore, the performance of the HMM-based line classifier [23] is also shown on the same plot. It can be noticed that the proposed Gaussian classification approach combined with slip-KS is better in performance than the previous HMM approach. Furthermore, the proposed approach has the advantage that it is able to instantly classify a given gaze point measurement after training the classifier, whereas the HMM approach, as it is implemented in [23], requires data from the entire page before the (Viterbi-based) classification algorithm can start.

Fig. 10(b) shows the prediction accuracy of the *line-by-line* classification approach. From this figure, the Gaussian-based classifier and the SVM-based classifier on the slip-KS approaches both outperform the HMM approach.

It can be observed that the performance of the Gaussian classifier, just with the KF outputs, is very poor. It is due to the fact that the performance of the Kalman filter is in such a way that the variance (particularly, in the y -direction) is not that reduced after filtering. This can be observed from Fig. 7. When the smoothing is applied, the variance significantly reduced, resulting in reduced overlap of the line distribution and much-improved classification performance.

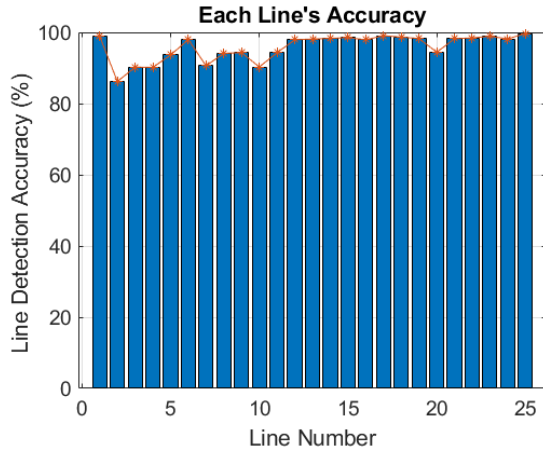


Fig. 11. Demonstration of eye-tracker bias. The eye-tracker is placed at the bottom of the screen (see Fig. 9) as such the classification accuracy can be seen to increase with the line number (top line on the screen is denoted 1 and the line number increases from top to bottom).

TABLE I
ACCURACY COMPARISON

	Mean	Standard Deviation
SVM + slip-KS (line-by-line)	97.92%	4.63%
Gaussian + slip-KS (line-by-line)	95.74%	5.68%
Gaussian + slip-KS (point-by-point)	89.15%	6.79%
Gaussian + slip-KF (line-by-line)	69.38%	27.73%
Gaussian + slip-KF (point-by-point)	65.21%	21.15%
HMM (page-by-page)	87.43%	4.03%

Fig. 11 shows the averaged accuracy of the proposed classification scheme on a line-by-line basis. It can be noticed that, except for the first line, the line classification accuracy increases from top lines toward the bottom ones. The placement of the eye-tracker (at the bottom of the screen) influences the above pattern in accuracy. The outlier in the first line is probably due to classification bias for initial line observation.

Table I shows the performance comparison of different algorithms analyzed in this article. In order to compare the performance against the previous approach in [23], the same dataset as in [23] was used for Table I. Compared to the classification based on slip-KF, the slip-KS-based classifier has a significantly higher mean accuracy and a smaller standard deviation which shows the benefit of data preprocessing using the slip-KS compared to that of slip-KF. Another advantage of the proposed approach is that the data does not need to be in a specific order, which means the accuracy will not be affected if the observation is measured while reading an arbitrary line of text on the screen. The average accuracy of the point-by-point classification based on KS is up to 89.15%. Moreover, for the line-by-line classification approach, the average accuracy is much higher than the point-by-point, which is up to 95.74%, much higher than the HMM approach proposed in [23]. Last but not least, the SVM-based approach reaches the highest accuracy 97.92% with the standard deviation of 4.63%.

The data collection procedure described in Section IV-A is repeated for six additional participants with the exception that, this time, each participant read ten pages. Fig. 12 shows

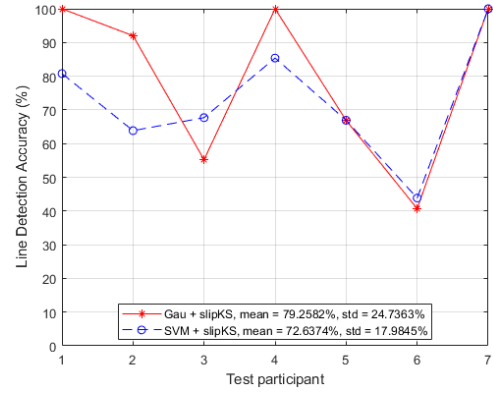


Fig. 12. Classification performance for all seven participants.

the comparison of classification accuracy for all seven participants. Initial experimentation with data from all seven subjects showed great variation in performance. As a result, it was decided to combine two adjacent lines together before the classification analysis. It is suspected that the observed variation might have been based on whether a participant is a native speaker who is fluent in English or not. This needs to be analyzed in detail using a separate study and is left as future work. As for the eye-tracking while reading, the Gaussian discriminant classifier shows a higher accuracy of 79.3% compared to the SVM accuracy at 72%. The SlipKS approach proved to be an important component for both classifiers.

D. Real-Time Analysis

In this section, some details are provided to understand the different algorithms analyzed in this article for their performance for real-time implementation.

It is reported that the average reading speed of a native English speaker is 4.06 words per second; for fast readers, the reading rate is 7.66 words per second [39]. In order to be used in real-life applications, the response time of the reading line classification algorithm should be less than the time it takes to finish reading a line. In the experimental setup of this article, there were 150 characters per line. According to statistics provided in [40], the average length of words in English is 4.7 characters. Based on these, the baseline latency of the experiment is defined as

$$\begin{aligned}
 T(\text{baseline}) &= \frac{\text{characters per line}}{\text{length of words} \times \text{reading rate}} \\
 &= \frac{150}{4.7 \times 4.06} = 7.86 \text{ sec.}
 \end{aligned}$$

For each algorithm, the latency is computed as

$$T(\text{algorithm}) = T_p + T_f + T_c \quad (14)$$

where T_p is the pre-processing time (SlipKS), T_f is the time taken for feature extraction (SVM only), and T_c is the time taken for classification. Table II shows the baseline latencies as well as the latencies of each algorithm considered in this article on a personal computer (i5-8500 CPU). It can be noticed that latencies of all the approaches considered in this

TABLE II
LATENCY ANALYSIS (LINE-BASED)

Classification approach	Average Latency (seconds)
$T(\text{baseline})$ - normal reading	7.86
$T(\text{baseline})$ - fast reading	4.166
$T(\text{SVM} + \text{slip-KS (line-by-line)})$	1.844
$T(\text{Gaussian} + \text{slip-KS (line-by-line)})$	0.034
$T(\text{Gaussian} + \text{slip-KF (line-by-line)})$	0.033

article are less than that of the two baselines. Furthermore, the Gaussian-based classifiers show significantly low latency compared to other classification approaches.

V. CONCLUSION AND DISCUSSIONS

In this article, a novel slip-KS based classification approach was developed to predict lines being read, on a real-time, point-by-point, or line-by-line basis, while reading. The proposed approach is presented as a novel slip-KS, which can detect the line returns and obtain less-variant measurements corresponding to each line being read. With the Kalman smoothed estimates, two different classification approaches, one based on the support vector machines (SVM) and the other bases on Gaussian discriminants are employed to classify smoothed gaze observations into possible text lines. Both of the proposed classification approaches can be used to predict the line being read in real-time. Both the SVM classifier and the Gaussian discriminant classifier show significant improvement in prediction accuracy compared to the previous approach and the SVM classifier performs best on the line-by-line classification. However, the disadvantage of the SVM classifier is that it cannot work in a point-by-point fashion. Both of the proposed approaches are less reliant on the continuity of lines while reading, making them more attractive in practical applications.

The present work opens several avenues for future research. Testing the proposed algorithm using data from multiple and diverse participants (e.g., different ages, different educational backgrounds, gender, different native languages) is one of them. Further, the proposed algorithm needs to be tested on more realistic reading settings (different head/body positions of the participants, different number of lines per page, multiple columns, etc.). Finally, the proposed approach needs to be demonstrated using the built-in webcam of a computer for suitability in wide ranging applications.

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